SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA5116
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Assessment Tool Weightage: **40%**

QUESTIONS: **Total Marks: 50**

Annexure A

Analytical Problem Solving

Code of Conduct:

I K.Vineela-192312123 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Vineela

Annexure A
Analytical Problem Solving

1. RSA Encryption Analysis
Given p=11, q=13

a. Compute n $\phi(n)$\
b. Choose e=7 and find the private key d
c. Encrypt the message M=9
d. Decrypt the ciphertext and verify the result
e. Analyze how changing e affects security

2. Block Cipher Mode Evaluation
Given plaintext blocks:
P=[1010,1010,1111,1010], Key = K=1100

a. Encrypt using ECB and CBC modes (Assume IV = 0000)
b. Compare the ciphertext outputs
c. Analyze which mode hides patterns better and explain why

3. DES vs 3-DES vs Blowfish Performance Study
Given:

a. DES key size = 56 bits, time = 2 ms/block
b. 3-DES key size = 168 bits, time = 6 ms/block
c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

i.Calculate total encryption time for each algorithm
ii.Analyze trade-offs between security and performance
iii.Recommend the best algorithm for secure real-time communication

4. Diffie-Hellman Key Exchange Analysis
Given:

a. p=23, g=5
b. User A private key = 6
c. User B private key = 15
d. Compute public keys for A and B
e. Compute the shared secret key
f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
g. Suggest a secure method to prevent the attack

5. ECC vs RSA Comparison

Given:

a. RSA key size = 1024 bits
b. ECC key size = 160 bits (equivalent security)
c. Device processing capacity = limited (IoT device)
d. Analyze computational cost for both systems
e. Estimate which consumes less power and memory
f. Justify which cryptosystem is better for the given scenario with reasoning

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganize d steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CO2
ASSESSMENT TOOL-1

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CSA5116 - Cryptography and Network security

1) RSA Encryption Analysis, Given p=11, q=13.

a) Compute n ϕ(n)\
b) Choose e=7 and find the private key d
c) Encrypt the message M=9
d) Decrypt the ciphertext and verify the result
e) Analyze how changing e affects security.

RSA Encryption Analysis - Detailed Explanation :-

- RSA stands for Rivest Shamir Adleman. It is a public key cryptographic algorithm used to protect information during communication. RSA uses two keys a public key to encrypt messages and a private key to decrypt message.

Given values,

prime number p = 11
prime number q = 13
public exponent e = 7
original message M = 9

a) Compute n and ϕ(n) :-

- Step 1 : Calculate the modulus n

$$n = p \times q$$

$n = 11 \times 13$

$\boxed{n = 143}$

The value of n is used in both encryption and decryption

-step 2 : calculate Euler's totient function :-

Euler's totient function, $\phi(n)$, represents the number of positive integers less than n that are relatively prime to n.

$$\boxed{\phi(n) = (p-1)(q-1)}$$

$$\phi(n) = (11-1)(13-1)$$

$$= 10 \times 12$$

$$\boxed{\phi(n) = 120}$$

- $n = 143$ and $\phi(n) = 120$

b) Choose $e = 7$ and find the private key d :-

The public Exponent e must be relatively prime to $\phi(n)$. This means their greatest common divisor must 1.

$$\gcd(7, 120) = 1$$

-step-1 : Apply the private key formula :-

The private Exponent d is the modular multiplicative inverse of e.

$$e \times d \equiv 1 \pmod{\phi(n)}$$

$$7d \equiv 1 \pmod{120}$$

This means that when $7d$ is divided by 120, the remainder must be 1.

- step-2 : Find d :-

Try d = 103

$7 \times 103 = 721$

$721 = 120 \times 6 + 1$

$d = 103$

The keys are :-

- Public key : $(e, n) = (7, 143)$
- private key : $(d, n) = (103, 143)$

c) Encrypt the message $M = 9$:-

Encryption converts the original message into cipher text using the public key.

The RSA Encryption formula is :-

$$\boxed{C = M^e \bmod n}$$

$C = 9^7 \bmod 143$

$9^2 = 81$

$9^4 = (81)^2 = 6561$

$6561 \bmod 143 = 126$

$9^4 = 126 \pmod{143}$

$\rightarrow 7 = 4+2+1$

$9^7 = 9^4 \times 9^2 \times 9$

$C = (126 \times 81 \times 9) \bmod 143$

$\rightarrow 126 \times 81 = 10206$

$10206 \bmod 143 = 53$

$53 \times 9 = 477$

$477 \bmod 143 = 48$

The encrypted message is :-

$$C = 48$$

The original message 9 has been converted into ciphertext 48. The ciphertext can be transmitted to the receiver, who can recover the original message using private key.

d). Decrypt the ciphertext and verify the result :-

Decryption converts ciphertext back into original message using the private key.

$$M = c^d \bmod n$$

$$M = 48^{103} \bmod 143$$

- step 1 : Express the exponent 103 :-

$$103 = 64 + 32 + 4 + 2 + 1$$

$$48^{103} = 48^{64} \times 48^{32} \times 48^{4} \times 48^{2} \times 48^{1} \text{ (remainder)}$$

$$M = 113 \times 16 \times 113 \times 16 \times 48 \pmod{143}$$

$$M = 9$$

- Original message : 9
- Encrypted mes ciphertext : 48
- Decrypted message : 9

Since the decrypted message is equal to original message, the RSA process is verified.

eg Analyze how changing e affects security :-

The public exponent e is an important parameter in RSA. It determines the mathematical operation used during encryption.

$$1 < e < \phi(n)$$

$$\boxed{\gcd(e, \phi(n)) = 1}$$

$\phi(n) = 120$, so $e = 7$ is valid

Effect of choosing small exponent :-

A small exponent generally makes encryption faster because fewer multiplication steps are needed. ($e=7$) is relatively small and valid for this example.

Effect of choosing a common practical exponent :-

Modern RSA implementations commonly use :-

$$\boxed{e = 65537}$$

This value is popular because it provides efficient encryption while avoiding some issues associated with extremely small exponents.

$$n = 143$$

$$e = 65537$$

$$e < \phi(n) = 120$$

Effect of choosing an Invalid exponent :-

Consider, $e = 10$

$$\gcd(10, 120) = 10$$

Conclusion :-

RSA demonstrates how public key cryptography enables secure communication using two different keys. Finally, changing the public exponent affects validity and computational efficiency, but a larger exponent alone does not guarantee better security.

4) Define-Hellman key exchange Analysis, given.

a). P = 23 , g = 5
b). User A private key = 6
c). User B private key = 15
d). Compute public keys for A and B
e). Compute the shared secret key
f). Demonstrate how a man in the middle attacker could alter the exchange.
g). Suggest a secure method to prevent the attack.

Diffie-Hellman (DH) is a cryptographic algorithm that allows two users to establish a shared secret key over a public communication channel. The shared key can than be used for secure communication.

Given values :-

prime number : P = 23

Generator : g = 5

User A's private key : a = 6

User B's private key : b = 15

a) Given Parameters :-

$P = 23$

$g = 5$

Both users know these values before starting the key exchange.

The private keys must remain secret throughout the exchange.

b) Compute the public key for user A :-

User A calculates the public key for user A.

$$A = g^a \bmod p$$

$$A = 5^6 \bmod 23$$

$$\rightarrow 5^2 = 25 = 2 \pmod{23}$$

$$5^4 = 2^2 = 4 \pmod{23}$$

$$5^6 = 5^4 \times 5^2$$

$$A = (4 \times 2) \bmod 23$$

$$\boxed{A = 8}$$

- User A's public key is 8.

c) Compute the public key for user B :-

User B calculates the public key using the formula.

$$B = g^b \bmod p$$

$$B = 5^{15} \bmod 23$$

$$5^2 \bmod 23 = 2$$

$5^4 \bmod 23 = 4$

$5^8 \bmod 23 = 16$

Since, $15 = 8+4+2+1$

$5^{15} = 16 \times 4 \times 2 \times 5 \pmod{23}$

$16 \times 4 = 64 \equiv 18 \pmod{23}$

$18 \times 2 = 36 = 13 \pmod{23}$

$13 \times 5 = 65 \equiv 19 \pmod{23}$

Therefore,

$$B = 19$$

- User B's public key is 9.

d) compute the shared secret key :-

Both user's exchange their public keys over the communication channel. User A receives B's public key, and User B receives A's public key.

For User A :-

$$K_A = B^a \bmod p$$

$K_A = 19^6 \bmod 23$

$\rightarrow 19^2 = 361 \equiv 16 \pmod{23}$

$19^4 = 16^2 = 256 = 3 \pmod{23}$

$19^6 = 19^4 \times 19^2$

$$K_A = (3 \times 16) \bmod 23$$

$$K_A = 48 \bmod 23$$

$$\boxed{K_A = 2}$$

For user B :-

$$K_B = 8^{15} \bmod 23$$

$$\rightarrow 8^2 \bmod 23 = 18$$

$$8^4 \bmod 23 = 18^2 \bmod 23 = 2$$

$$8^8 \bmod 23 = 2^2 \bmod 23 = 4$$

Since, $15 = 8 + 4 + 2 + 1$

$$8^{15} = 4 \times 2 \times 18 \times 8 \pmod{23}$$

$$4 \times 2 = 8$$

$$8 \times 18 = 144 = 16 \pmod{23}$$

$$6 \times 8 = 48 = 2 \pmod{23}$$

$$\boxed{K_B = 2}$$

- Both users calculate the same shared secret.

$$\boxed{K_A = K_B = 2}$$

e). Demonstrate a Man in the Middle (MITM) Attack:-

User A (Private key = 6)

↓↑

Attacker (Mallory)

↓↑

User B (Private key = 15)

f) Example of MITM attack :-

For illustration, suppose mallory chooses private value $m = 3$,

$$X = g^m \bmod p$$

$$X = 5^3 \bmod 23$$

$$X = 125 \bmod 23$$

$$\boxed{X = 10}$$

User A calculates :-

$$K_{AM} = 10^6 \bmod 23$$

$$= (18 \times 8) \bmod 23$$

$$\boxed{K_{AM} = 6}$$

User B calculates :-

$$K_{BM} = 10^{15} \bmod 23 = (2 \times 18 \times 8 \times 10) \bmod 23$$

$$\boxed{K_{BM} = 6}$$

g) Suggest a secure method to prevent the attack :-

The primary solution is to authenticate the exchanged public keys.

- Digital signatures
- Digital certificates and PKI
- Use authenticated protocols
- Verify public key fingerprints.

For real-world security, use a well reviewed authenticated key exchang protocol.